Using R to analyze Mario-Kart Wii sales on eBay to see how the inclusion of different variables influences final sale price

Objective: The purpose of this project is to examine how different variables influence the final sale price of Mario-Kart Wii on eBay through 3 specific questions:

1. Does the inclusion of wheel(s) (the wheel is an accessory for the wii controller to make it look/feel like you are really turning a car wheel) significantly increase the final sale price?
2. Does including an original photo of the product (not a stock photo) significantly increase the final sale price?
3. Does combining both original photos and the inclusion of wheel(s) provide a finale sale price higher than what each would provide individually?

Variables: There are 13 variables included in this dataset such as auction ID assigned by eBay (id), auction length in days (duration), number of bids (n_bids), game condition either new or used (cond), start price of the auction (start_pr), shipping price (ship_pr), total price which is the auction price plus the shipping price (total_pr), shipping speed/method (ship_sp), seller's rating on eBay which is the number of positive ratings minus the negative ratings (seller_rate), whether the listing included a stock photo or not (stock_photo), the number of wheels (accessories for the wii controller to make it seem like you are turning) included (wheels), and the title of the auctions (title).

Data Preperation/Cleaning

```
> mariokart <- read.csv("C:\\Users\\brian\\Downloads\\mariokart.
csv")
> clean_mk <- subset(mariokart, total_pr < 100)
> clean_mk$wheels <- ifelse(clean_mk$wheels > 0, "yes", "no")
```

I imported data from the openintro library and I observed that there were outliers so I set th e max selling price to $100 filtering out observations of a higher final sale price in a new dat

aset. Afterwards, I recoded the wheels column in this new dataset to either display a yes fo r having more than 0 wheels or no for having 0 wheels.

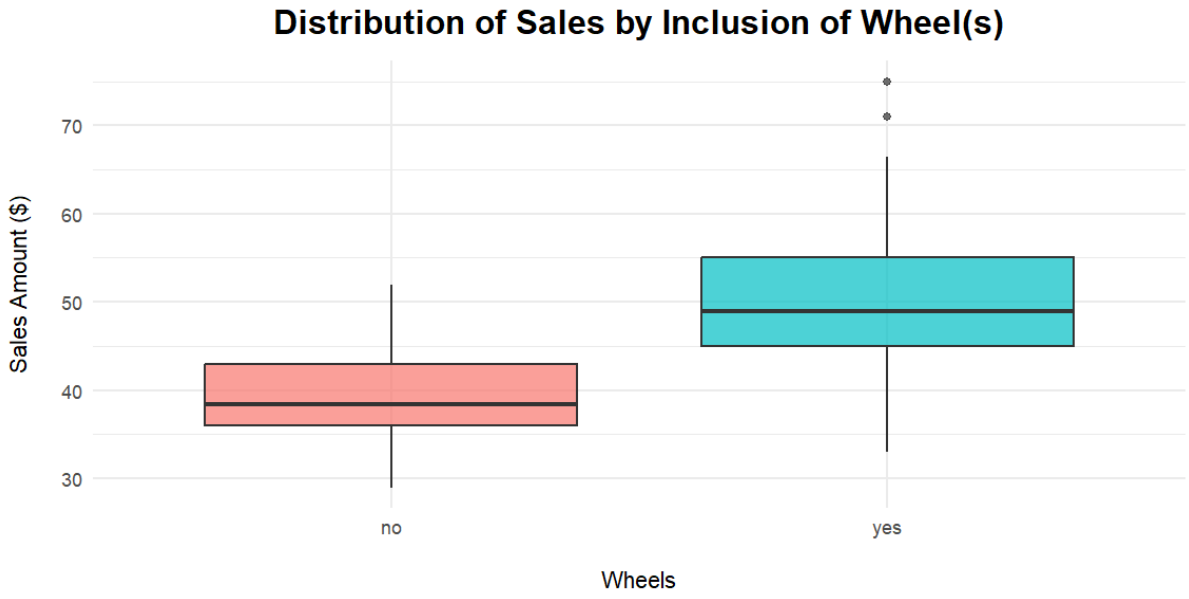
Statistics

```
> library(tidyverse)
> clean_mk %>%
+   group_by(wheels) %>%
+   summarize(Mean_Price = mean(total_pr, na.rm = T), SD_Price =
sd(total_pr, na.rm = T), Sample_Size = n())
# A tibble: 2 × 4
  wheels Mean_Price SD_Price Sample_Size
  <chr>       <dbl>    <dbl>       <int>
1 no           38.9     5.77          36
2 yes          50.4     8.17         105

> library(tidyverse)
> clean_mk %>%
+   group_by(stock_photo) %>%
+   summarize(Mean_Price = mean(total_pr, na.rm = T), SD_Price =
sd(total_pr, na.rm = T), Sample_Size = n())
# A tibble: 2 × 4
  stock_photo Mean_Price SD_Price Sample_Size
  <fct>            <dbl>    <dbl>       <int>
1 no                44.3     7.25          36
2 yes               48.5     9.47         105
```

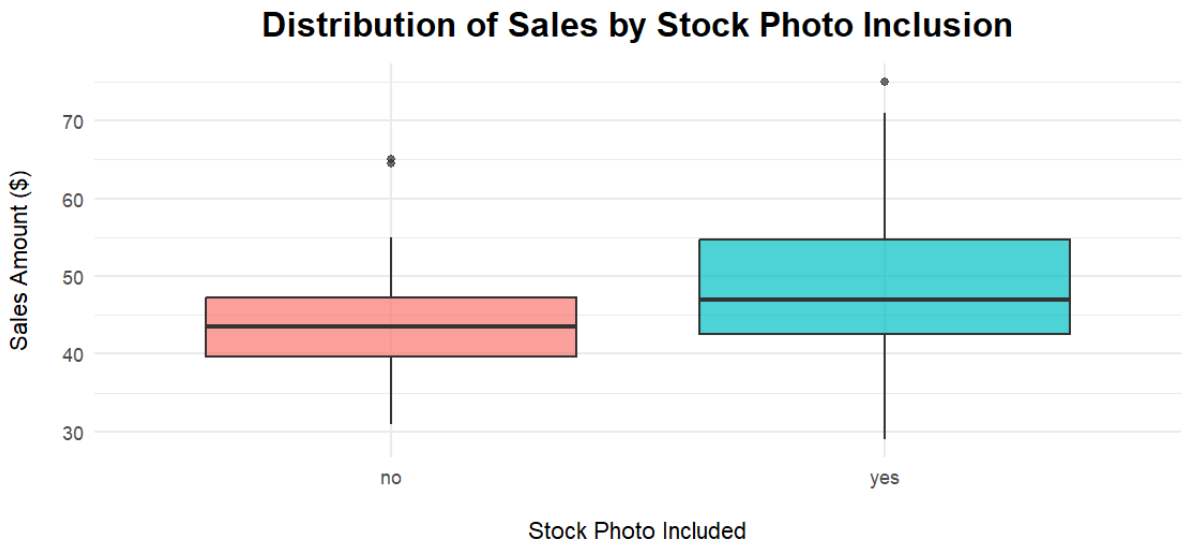
Here are statistics showing the mean price, standard deviation price, and sample size of the dataset based on the wheels variable and the stock photo variable. From this data it is apparent that the average final sale price is $11.50 higher when including at least 1 wheel than including no wheels. It is also important to know that the average final sale price is $4.20 higher when including stock photos than when not. The key takeaway is that although both individually increase final sale price, including wheel(s) greatly increases final sale price.

Visualizations




```
> library(tidyverse)
> ggplot(clean_mk, aes(x = wheels, y = total_pr, fill = wheels))
+
+   geom_boxplot(alpha = 0.7) +
+   theme_minimal() +
+   labs(title = "Distribution of Sales by Inclusion of Wheel(s)
", x = "Wheels", y = "Sales Amount ($)") +
+   theme(axis.title.y = element_text(margin = margin(r = 15)),
+         axis.title.x = element_text(margin = margin(t = 15)),
+         plot.title = element_text(hjust = 0.5, size = 16, face
= "bold", margin = margin(b=10)),
+         legend.position = "none")
```

From this graph it is observable that the median selling price for listings with wheel(s) is $49 whereas listings with no wheels have a median of about $38. It is also worth noting that the entire “yes” box sits higher than the entire “no” box, this means that most listings with whe els sold for more money than most listings without wheels.




```
> library(tidyverse)
> ggplot(clean_mk, aes(x = stock_photo, y = total_pr, fill = sto
ck_photo)) +
+   geom_boxplot(alpha = 0.7) +
+   theme_minimal() +
+   labs(title = "Distribution of Sales by Stock Photo Inclusion
", x = "Stock Photo Included", y = "Sales Amount ($)") +
+   theme(axis.title.y = element_text(margin = margin(r = 15)),
+         axis.title.x = element_text(margin = margin(t = 15)),
+         plot.title = element_text(hjust = 0.5, size = 16, face
= "bold", margin = margin(b=10)),
+         legend.position = "none")
```

From this graph it is observable that the median selling price for listings with a stock photo i s $47 whereas listings with no stock photo have a median of about $44. Although the differ ence of inclusion of stock photo is not as drastic as wheels, there is still a noticeable increa se of difference in the final sales for inclusion of stock photos.

Normality Tests

```
> shapiro.test(clean_mk$total_pr[clean_mk$wheels == "yes"])

        Shapiro-Wilk normality test

data:  clean_mk$total_pr[clean_mk$wheels == "yes"]
W = 0.97329, p-value = 0.03214

> shapiro.test(clean_mk$total_pr[clean_mk$wheels == "no"])

        Shapiro-Wilk normality test
```

```
data:  clean_mk$total_pr[clean_mk$wheels == "no"]
W = 0.9726, p-value = 0.5008
```

I ran shapiro tests to see if each group of listings (yes=has wheel, no=no wheel) follows nor mal distribution. Listings with wheels do not adhere to strict normality, but listings without wheels do. However, due to the central limit theorem the sample size provides reliability for the data to be used for a t test.

```
> shapiro.test(clean_mk$total_pr[clean_mk$stock_photo == "yes"])

        Shapiro-Wilk normality test

data:  clean_mk$total_pr[clean_mk$stock_photo == "yes"]
W = 0.9886, p-value = 0.5175

> shapiro.test(clean_mk$total_pr[clean_mk$stock_photo == "no"])

        Shapiro-Wilk normality test

data:  clean_mk$total_pr[clean_mk$stock_photo == "no"]
W = 0.92138, p-value = 0.01382
```

I ran shapiro tests to see if each group of listings (yes=has stock photo, no=no stock photo) follows normal distribution. Listings with stock photos do not violate strict normality, but lis tings without stock photos do. However much like the wheels, due to the central limit theor em the sample size provides reliability for the data to be used for a t test.

Tests

```
> t.test(total_pr ~ wheels, data = clean_mk)

        Welch Two Sample t-test

data:  total_pr by wheels
t = -9.158, df = 86.002, p-value = 2.345e-14
alternative hypothesis: true difference in means between group n
o and group yes is not equal to 0
95 percent confidence interval:
 -13.930073  -8.961086
sample estimates:
 mean in group no mean in group yes
         38.90861          50.35419
```

Due to the p-value being smaller than 0.05, we are 95% confident that there is statistically significant evidence that including wheel(s) in the listing of Mario-Kart Wii on eBay increase s the final sale price by $8.96-$13.93.

```
> t.test(total_pr ~ stock_photo, data = clean_mk)

        Welch Two Sample t-test

data:  total_pr by stock_photo
t = -2.7416, df = 78.872, p-value = 0.007562
alternative hypothesis: true difference in means between group n
o and group yes is not equal to 0
95 percent confidence interval:
 -7.196157 -1.142161
sample estimates:
 mean in group no mean in group yes
         44.32722           48.49638
```

Due to the p-value being smaller than 0.05, we are 95% confident that there is statistically significant evidence that including a stock photo in the listing of Mario-Kart Wii on eBay incr eases the final sale price by $1.14-$7.20.

Conclusion

```
> anova_result<- aov(total_pr ~ wheels * stock_photo, data = clean_mk)
> summary(anova_result)
                    Df Sum Sq Mean Sq F value   Pr(>F)
wheels               1   3512    3512  63.141 6.32e-13 ***
stock_photo          1    392     392   7.054  0.00885 **
wheels:stock_photo   1    104     104   1.867  0.17401
Residuals          137   7620      56
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

In conclusion, from the results of the anova test it is clear that although independently the variables yield an increase in final sale price which is proven by their p-values being less than 0.05, there is no statistically significant evidence to prove that when combined a great bonus is provided. This is proven by the anova test’s p-value of wheels and stock photo’s interaction term being greater than 0.05.

Although in tandem they do not excel, it is still important to note that there is statistically significant evidence that including at least one wheel will significantly increase the final sale price and that including a stock photo will slightly increase the final sale price.

For the sake of my future work, one critical limitation that I realized was the omittance of certain variables from this dataset. Variables like feedback ratings and shipping costs were not included in my analysis, but in the future I should take account of all variables that can influence what it is I am researching. To add on to this, it is vital that going forward I use datasets that contain more observations. A larger dataset would ensure that my research can be more generalized and applicable to more video game accessories and more marketplaces other than just eBay.